Daniel Zhu

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EDUCATION

University of Colorado Boulder, Boulder, CO — Expected May 2027

- B.S. in Computer Science
- GPA: 3.9/4.0 | Honors and Organizations: CU Robotics, CU Mahjong
- Relevant Coursework: Algorithms, Machine Learning, Neural Networks & Deep Learning, AI Engineering, Deep Reinforcement Learning, Advanced Robotics, Human-Robot Interaction

SKILLS

Python, C++, PyTorch, Git, Docker, Linux, ROS2, Arduino, Soldering, CAD, 3D Printing, Beekeeping

EXPERIENCE

Software Lead, *CU Robotics,* Boulder, CO — Aug 2023 - Present

- CU Robotics Club builds mobile projectile-firing robots to compete in the ARC Robotics (formerly RoboMasters) competition. https://www.curobotics.net
- Trained, benchmarked, and deployed a YOLO26 model and a custom icon detector on an NVIDIA Jetson Orin AGX to recognize plates in an image at 160+fps with TensorRT optimization to detect locations of enemy robots.
- Created our custom armor plate dataset comprised of 500+ images of our robots with the different possible plate icons (5) and colors (3) and the plates by themselves, unattached to prevent overfitting to our robots. Developed a SAM3 Label Studio backend to help auto-annotate our data.
- Led a sub-team working on improving our autonomous robot’s pathing and behavior by developing a heatmap that tracks risk in the environment used as a heuristic to our A* path planner to incentivize our robot to move around high-risk areas.

Research Assistant, *Image and Video Computing Group,* Boulder, CO — Jul 2026 – Present

- Working under Dr. Danna Gurari on creating “focus ambiguous” questions to challenge modern-day Vision Language Models (VLMs) and encourage their development.
- Developing a focus ambiguity video dataset by giving VLMs personas, e.g., blind/low vision, to generate realistic and useful questions, along with verifying with the respective groups to see if they’re accurate.

Research Assistant, *Peleg Lab,* Boulder, CO — May 2025 – Aug 2025

- Worked under Dr. Danielle Chase to analyze the behavior of honeybee swarms on how they form and adapt to the environment to gather insights into swarm intelligence. Presentation
- Designed, built, and implemented an experimental setup with the team that successfully captured honeybee swarm formation in the natural environment.
- Created a program to track the rotation of motors to detect deviations in rotation speed over long periods of time to detect defects. Another, which automates camera calibration and synchronization between two stereo cameras. Camera-Localization
- Designed and 3d printed custom GoPro mounts and adapters that attached them to our experimental setup.

OTHER EXPERIENCE

Senior Thesis: “Bridging the LLM Prompting Gap” — Aug 2026 - Present

- Advised by Dr. Bradley Hayes. As shown time and again, prompting can have a major impact on LLM outputs. Experienced users know how to effectively prompt, provide context, and guide LLMs toward their desired outputs. The common person often lacks this knowledge, creating a “prompting” gap. We aim to create an intervention system to improve prompts, align goals, and reduce brain drain for the common people.

CU Mahjong Club President — Sep 2024 - Present

- Founded and run the Mahjong Club, with over 200 Discord members! Together with my team, we organize lessons/tutorials, manage finances, and collaborate with other clubs, departments, and universities to host events where people can come and make friends through Mahjong. (@mahjongclub_cu)